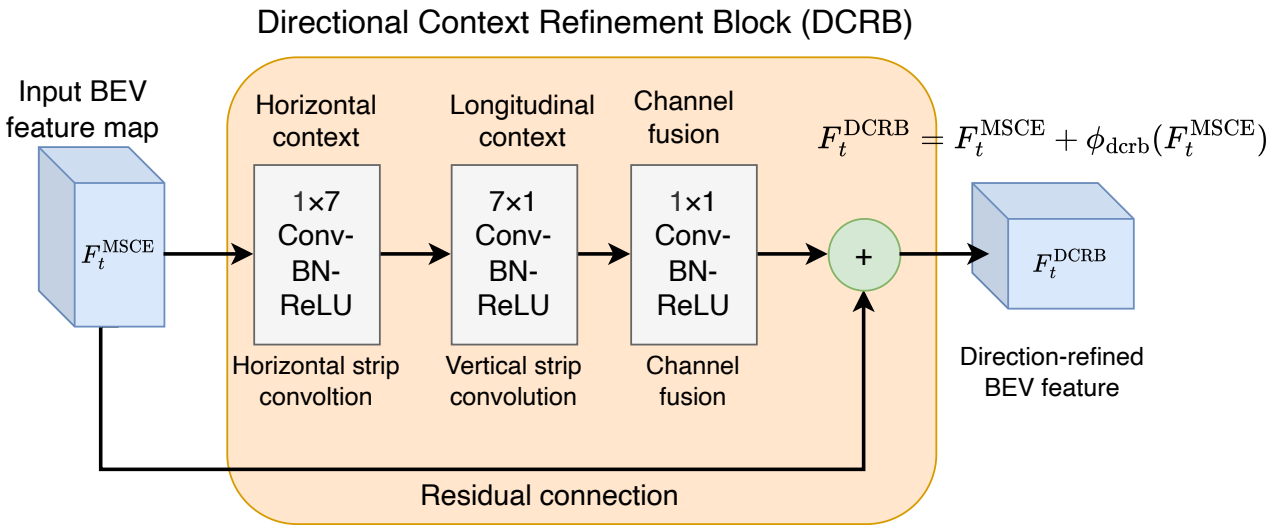
Directional Context Refinement Block (DCRB)
Input BEV feature map
$F_t^{\mathrm{MSCE}}$
Horizontal context
1×7 Conv-BN-ReLU
Horizontal strip convoltion
Longitudinal context
7×1 Conv-BN-ReLU
Vertical strip convolution
Channel fusion
1×1 Conv-BN-ReLU
Channel fusion
$F_t^{\mathrm{DCRB}} = F_t^{\mathrm{MSCE}} + \phi_{\mathrm{dcrb}}(F_t^{\mathrm{MSCE}})$
+
$F_t^{\mathrm{DCRB}}$
Direction-refined BEV feature
Residual connection